The Exoplanet Radius–Response Atlas

A candidate Hertzsprung–Russell-like organizing diagram for exoplanets

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Abstract

A Hertzsprung–Russell-like diagram for exoplanets should organize the population into physically interpretable sequences, transitions, and outliers rather than merely maximize a two-variable correlation. We analyzed a 24 August 2026 snapshot of the NASA Exoplanet Archive Planetary Systems table, screening 190 two-parameter projections, 1,365 four-parameter spaces, and 165 radius response surfaces. After precision cuts, the principal sample contains 1,217 transiting planets with measured masses, radii, and incident fluxes; 1,141 also have host metallicities. The best non-tautological organizer is a mass–irradiation control plane in which planet radius is the response field. A collapsed atlas based on mass-corrected irradiation resolves five empirical zones: high-forcing compact planets, moderate-forcing sub-Neptunes, a cool giant shelf, warm compact giants, and hot inflated giants. The underlying response field changes character across a broad transition at approximately 50–100 Earth masses: the radius–mass slope collapses, the irradiation response changes from negative to positive, and the metallicity response changes from a localized positive association at 10–30 Earth masses to a negative association among giants. A two-branch mass–flux–metallicity model reduces host-grouped held-out radius error by 37.6% relative to a single global plane. We state quantitative falsification tests for the transition mass, the two irradiation slopes, the metallicity sign reversal, the sufficiency of the collapsed giant-planet coordinate, and future out-of-sample performance. The atlas is proposed as an empirical population diagram, not as a finalized taxonomy or a causal evolutionary model.

1 Introduction

The Hertzsprung–Russell (H–R) diagram is useful because it exposes sequences, turnoffs, evolutionary states, and diagnostic outliers in a compact observational plane. Exoplanet population studies have produced several important diagrams, including mass–radius relations, period–radius distributions, radius-gap maps, and irradiation–radius relations [2, 5, 6, 11]. Each captures part of the population structure, but no single diagram has become a comparably general organizer of planet structure and environmental response.

An exoplanet analogue need not use the same kinds of axes as the stellar H–R diagram. Planetary radius depends simultaneously on bulk mass, envelope fraction, composition, thermal state, irradiation history, and age. A useful diagram should therefore separate *control variables* from the *response field*. The analysis developed here identifies planet mass and bolometric incident flux as the most informative two-dimensional control plane, with radius as the displayed response and host metallicity as a composition-sensitive residual coordinate.

The most compact representation of this structure is the *Exoplanet Radius–Response Atlas*. It folds mass and irradiation into one horizontal coordinate while retaining planet radius vertically. The resulting chart reveals both the low-mass and giant-planet sequences, while the full mass–irradiation diagram explains the response turnover that creates those sequences.

2 Data and search methodology

2.1 Catalog and precision cuts

The source catalog is the NASA Exoplanet Archive Planetary Systems table [1], restricted to transiting planets and the archive-selected default solution for each object. The query returned 4,731 rows. Seven controversially flagged entries were removed before feature construction. The principal analysis sample comprises 1,217 planets satisfying all of the following conditions:

- measured planet mass and radius;
- fractional mass uncertainty no greater than 50%;
- fractional radius uncertainty no greater than 20%; and
- a measured or consistently derived bolometric incident flux.

Host metallicity is available for 1,141 of these planets. Incident flux was derived uniformly from stellar radius, effective temperature, stellar mass, and orbital period when those inputs were available; the archive flux value was used as a fallback. The analysis is therefore a study of the published mass-measured transiting sample, not an occurrence-corrected census.

2.2 Parameter-space screen

The search included direct and transformed quantities describing planet mass, radius, period, irradiation, equilibrium temperature, density, surface gravity, orbital scale, stellar properties, and composition. It comprised three complementary screens:

1. all 190 eligible two-parameter projections;
2. all 1,365 eligible four-parameter combinations; and
3. 165 response surfaces in which radius was modeled using every three-variable combination drawn from upstream controls.

Derived quantities containing planet radius were excluded from the radius-response search to prevent target leakage. Algebraically redundant clusterings—for example, combinations of mass, radius, density, gravity, and escape speed—were not treated as discoveries even when they produced visually compact groups. The primary criterion was whether a projection preserved non-tautological population structure and improved out-of-sample radius prediction.

2.3 Piecewise response model

The selected response family is

$$\log_{10}\left(\frac{R_p}{R_\oplus}\right) = a_j + b_{M,j}\log_{10}\left(\frac{M_p}{M_\oplus}\right) + b_{F,j}\log_{10}\left(\frac{F_\star}{F_\oplus}\right) + b_{Z,j}[\mathrm{Fe/H}], \tag{1}$$

where j denotes the low-mass or giant response branch. Candidate break masses were evaluated by information criteria, host bootstrap, measurement-error perturbation, and host-grouped prediction. The data support a broad transition band rather than a sharp universal boundary.

3 The Exoplanet Radius–Response Atlas

The collapsed coordinate is

$$\mathcal{I}_{0.4} = \left(\frac{F_\star}{F_\oplus}\right)\left(\frac{M_p}{M_\oplus}\right)^{-0.4}. \tag{2}$$

The exponent is an empirical approximation to the giant-planet inflation ridge: more massive giant planets require greater forcing to occupy the same radius sequence. Giant-only cross-validation favors exponents near 0.45–0.50; the rounded value 0.4 provides a stable and interpretable projection. Both axes of the atlas are logarithmic.

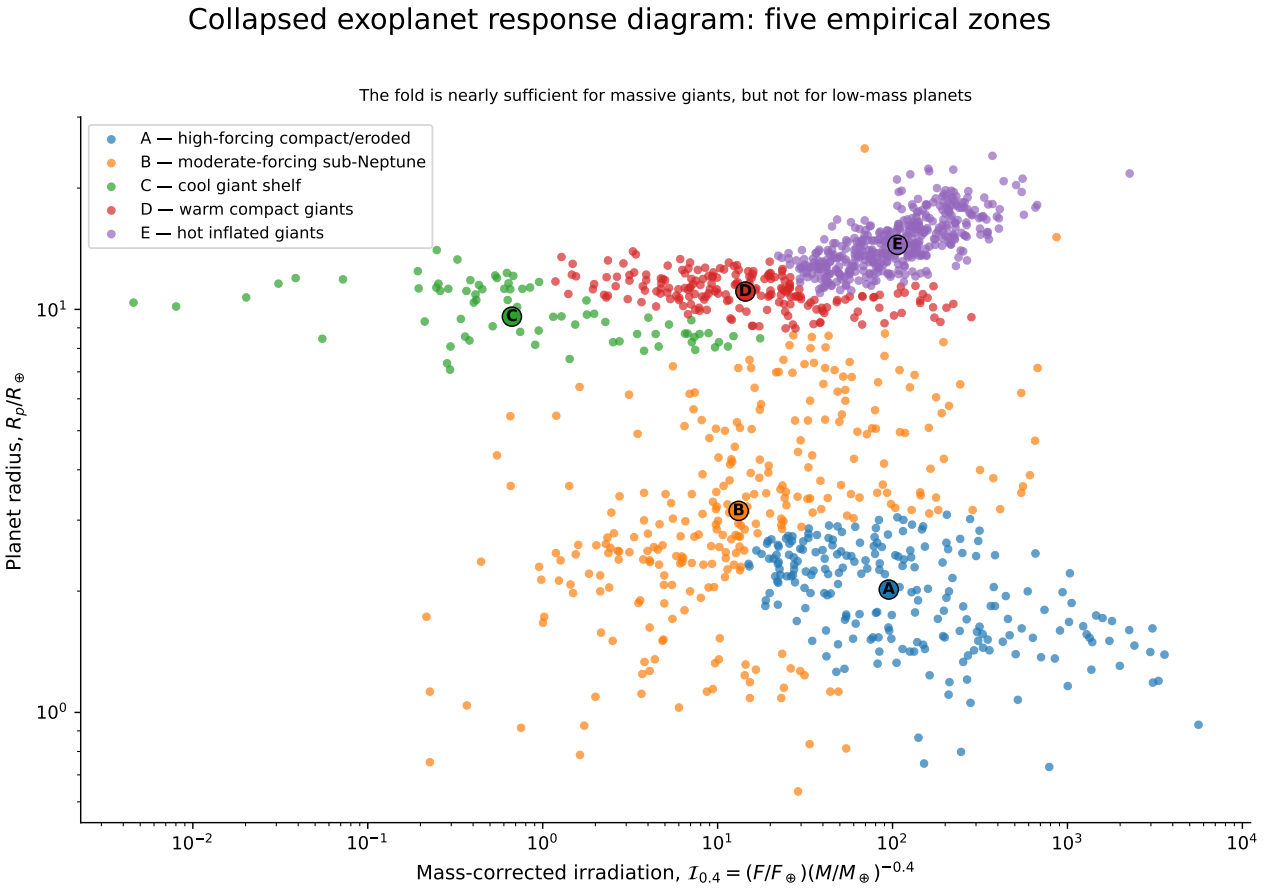


Figure 1: The Exoplanet Radius–Response Atlas. Planet radius is plotted against the mass-corrected irradiation coordinate defined in Eq. (2). Five empirical zones summarize the observed morphology. The fold is nearly sufficient for the giant inflation sequence but does not remove the independent importance of mass, envelope fraction, and composition among low-mass planets.

The atlas in Fig. 1 separates five empirical zones. Their descriptive medians are given in Table 1. These zones are population concentrations rather than hard class boundaries. Four- and five-component Gaussian-mixture descriptions are nearly tied by Bayesian information criterion, and the cool giant shelf is the least secure independent component. Host-bootstrap stability of the five-zone assignment has a median adjusted Rand index of 0.79, with a 5–95% range of 0.65–0.92.

Zone A contains compact planets under strong present-day forcing and is consistent with a mixture of rocky planets, stripped cores, and planets with thin surviving envelopes. Zone B contains the broader sub-Neptune population. Zones C–E trace the giant-planet sequence from weakly irradiated,

Table 1: Empirical zones in the collapsed atlas. Values are sample medians and should not be interpreted as defining boundaries.

Zone	Description	N	$M_p/M_\oplus$	$R_p/R_\oplus$	$F_\star/F_\oplus$	P (d)	ρ (g cm^{-3})
A	high-forcing compact/eroded	221	6.4	2.02	209.9	4.6	4.51
B	moderate-forcing sub-Neptune	286	10.7	3.16	31.8	10.7	2.03
C	cool giant shelf	74	196.5	9.59	7.5	78.6	0.91
D	warm compact giants	197	246.0	11.07	138.6	5.2	0.97
E	hot inflated giants	439	311.5	14.46	1063.2	3.4	0.55

approximately Jupiter-sized planets to strongly irradiated, inflated giants. The atlas is most effective as a morphology map; the full control plane is required to identify the response changes that produce the morphology.

4 The full mass–irradiation response field

Figure 2 displays the parent diagram: planet mass versus bolometric incident flux, with radius encoded by color. The broad transition between the low-mass and giant branches is centered near Saturn/sub-Saturn masses.

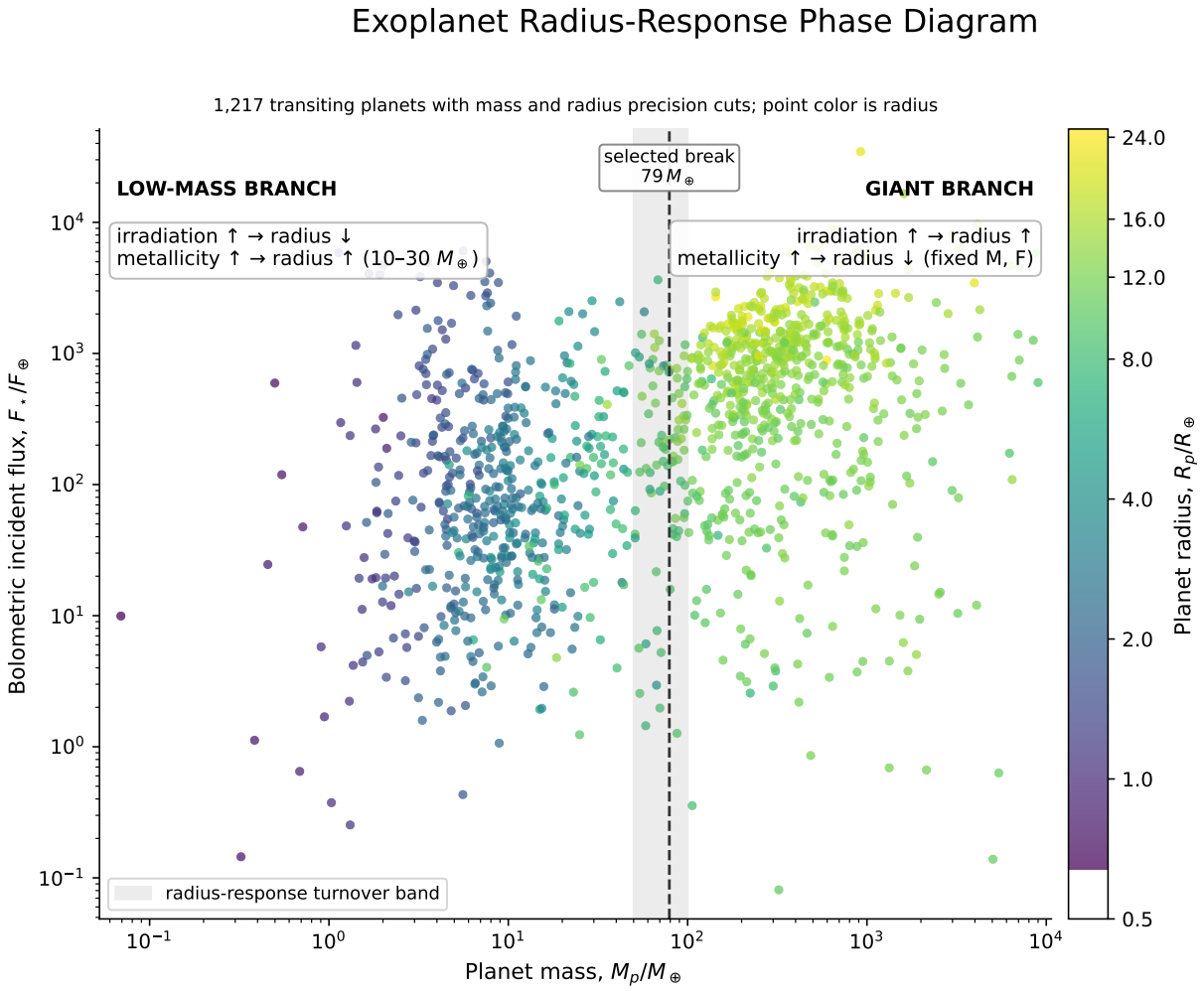


Figure 2: The Exoplanet Radius–Response Diagram. Planet mass and incident flux are the control coordinates; planet radius is the response field. The gray interval represents the broad response-turnover region supported by resampling, and the dashed line marks the selected break near 79 $M_\oplus$.

At a fixed break of 80 $M_\oplus$, the branch coefficients are listed in Table 2. Three changes occur across the transition:

1. the radius response to mass decreases from a steep positive slope to a value statistically consistent with zero;
2. the radius response to irradiation changes from negative to positive; and
3. the metallicity response changes from a localized positive association in the 10–30 $M_\oplus$ interval to a negative association among giant planets.

The coefficients have direct effect-size interpretations. On the low-mass branch, a factor of ten increase in incident flux corresponds to an 8.9% decrease in radius at fixed mass and metallicity. On the giant branch, the same flux increase corresponds to a 17.8% increase in radius. Within the 10–30 $M_\oplus$ subset, a +0.5 dex metallicity difference corresponds to a 25.5% larger radius at fixed mass and flux; above 80 $M_\oplus$, the corresponding change is an 8.5% smaller radius.

The moving-window fits in Fig. 3 show that these changes are gradual. The irradiation coefficient becomes securely positive near the Saturn-mass regime, the metallicity coefficient becomes negative at

Table 2: Host-clustered coefficients for Eq. (1). The 10–30 $M_\oplus$ row is a diagnostic subset of the low-mass branch.

Regime	N	b_M	p_M	b_F	p_F	b_Z	p_Z
$M_p < 80\,M_\oplus$	509	+0.527	1.9×10^{-163}	-0.040	3.4×10^{-6}	+0.048	0.186
10–30 $M_\oplus$	149	+0.657	1.3×10^{-15}	-0.039	0.048	+0.197	0.0045
$M_p \geq 80\,M_\oplus$	632	+0.005	0.531	+0.071	1.2×10^{-66}	-0.077	6.2×10^{-9}

a somewhat lower mass, and the mass coefficient approaches the giant-planet radius plateau over a broad mass interval. The term *response turnoff* is therefore more appropriate than a discrete class boundary.

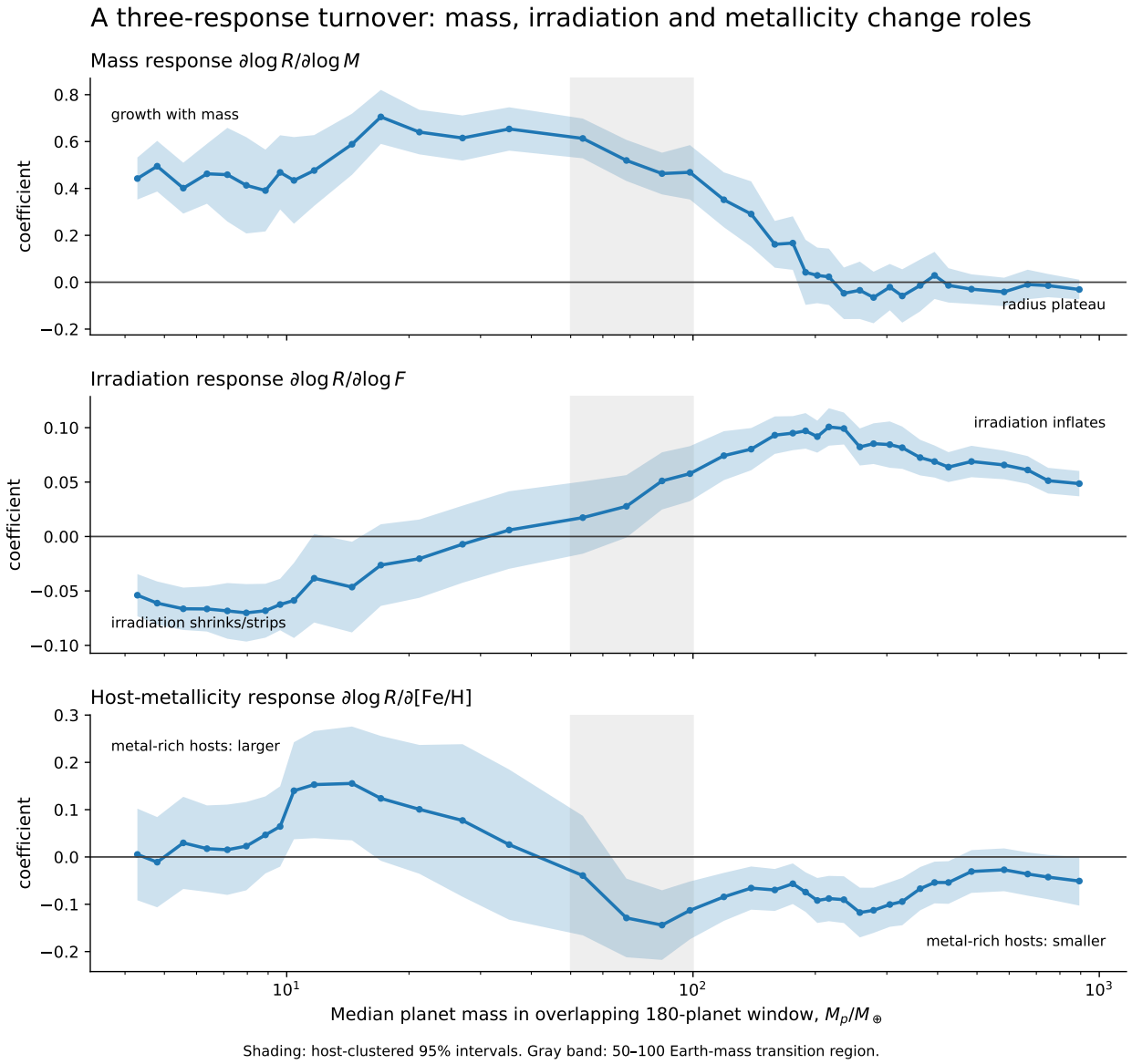


Figure 3: Mass-dependent response coefficients estimated in moving 180-planet windows. Shaded intervals are host-clustered 95% confidence intervals. The broad turnover consists of a collapse in the mass coefficient, an irradiation sign reversal, and a metallicity sign reversal.

5 Validation and robustness

5.1 Breakpoint stability

The selected break is not an exact physical constant. Host bootstrap resampling gives a median of 79.1 $M_\oplus$ and a 95% interval of 53.2–101.2 $M_\oplus$. Measurement-error perturbations give a median of 74.8 $M_\oplus$ and a 95% interval of 56.6–88.0 $M_\oplus$. These distributions are shown in Fig. 4.

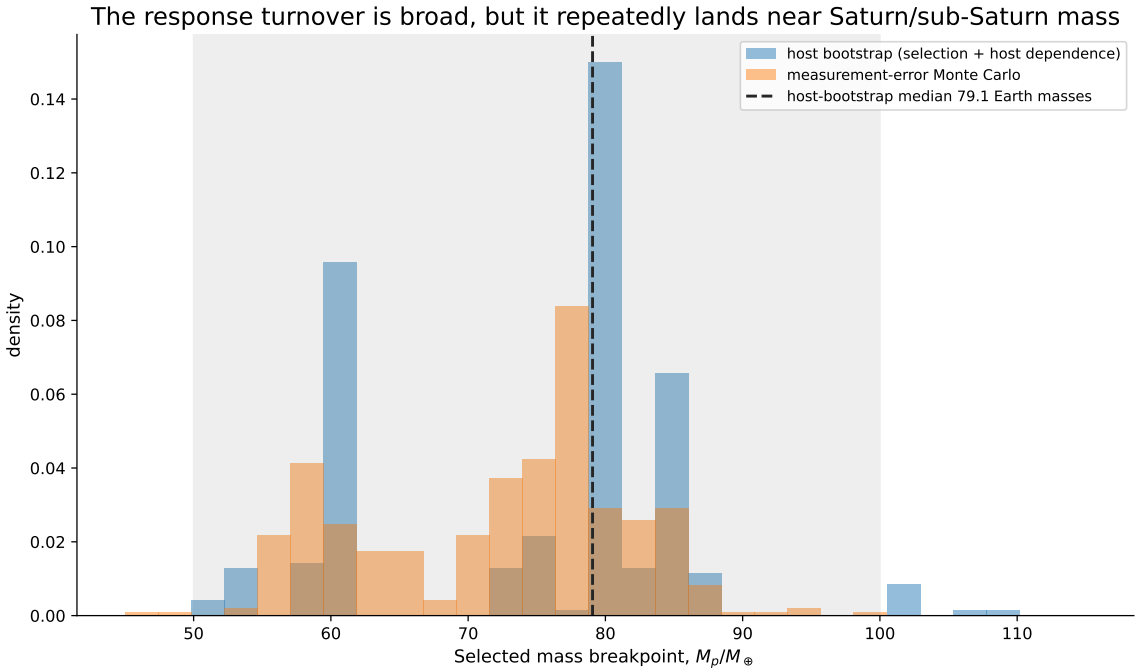


Figure 4: Breakpoint stability under host bootstrap resampling and measurement-error perturbation. Both analyses support a broad Saturn/sub-Saturn transition rather than a unique, sharply determined mass.

5.2 Out-of-sample prediction

One hundred host-grouped train/test splits were used to prevent planets around the same star from appearing in both training and test sets. The global mass–flux–metallicity model achieved a mean held-out radius RMSE of 0.1610 dex. The fixed-80 $M_\oplus$ two-branch model achieved 0.1005 dex, a mean reduction of 37.6%. When the break was reselected within each training set, the piecewise model outperformed the global model in every split; the median training-selected break was 82.6 $M_\oplus$.

The principal coefficient directions also persist in Kepler, K2, TESS, ground-discovered, radial-velocity-mass, FGK-host, and discovery-era subsets where the sample size is adequate. They remain under direct archive fluxes, additional stellar controls, facility and mass-method fixed effects, radius-error weighting, and Huber robust regression. A metallicity permutation test stratified by discovery facility and stellar-mass quintile gives $p = 0.00166$ for the observed low- versus high-mass metallicity-slope contrast.

5.3 Comparison with alternative diagrams

The mass–irradiation plane ranked first in a common projection benchmark for the latent state $(R_p, M_p, F_\star, [\mathrm{Fe/H}])$. Its mean information score was 0.858, compared with 0.853 for metallicity–radius,

0.848 for radius–irradiation, 0.830 for the collapsed atlas, and 0.802 for mass–radius. The benchmark combines neighborhood trustworthiness, reverse continuity, distance-rank preservation, and out-of-fold recovery of a common coarse partition. It is a comparative diagnostic rather than a likelihood.

The leading response surfaces are shown in Fig. 5. Several alternatives involving orbital geometry or stellar temperature are statistically close to the selected model. The claim therefore rests on the stability of the response topology and its predictive gain, not on a uniquely dominant set of covariates.

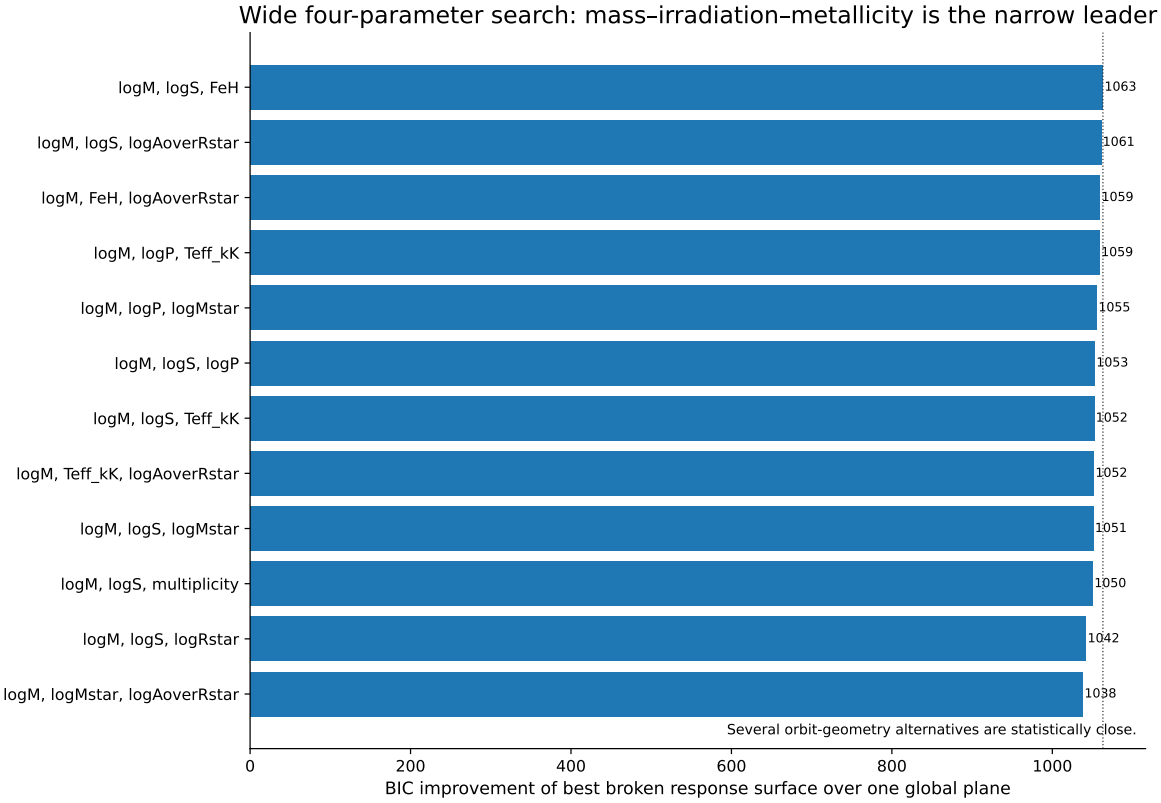


Figure 5: Leading four-variable radius response surfaces. The mass–irradiation–metallicity surface is the narrow information-criterion leader, while several orbital-geometry alternatives are close.

6 Falsifiable predictions

The atlas and the full response field imply the quantitative tests in Table 3. Predictions P1–P6 follow directly from the fitted response topology. P7 and P8 test whether the atlas also captures population structure beyond the regression coefficients.

Table 3: Falsifiable predictions derived from the Exoplanet Radius–Response Atlas.

ID	Prediction	Quantitative test	Failure criterion
P1	Low-mass irradiation response	In a homogeneous sample of approximately 5–30 $M_\oplus$ planets, the conditional coefficient $\partial \log R_p / \partial \log F_\star$ should be negative, with a target value near -0.04.	A common-pipeline sample yields a non-negative low-mass coefficient after controlling for mass and host composition.

Table 3: Falsifiable predictions derived from the Exoplanet Radius–Response Atlas (continued).

ID	Prediction	Quantitative test	Failure criterion
P2	Giant-planet irradiation response	Above approximately 150 $M_\oplus$, the conditional irradiation coefficient should be positive and near +0.07, corresponding to roughly 18% greater radius per decade in flux.	The giant-branch coefficient is consistent with zero or negative in a large homogeneous sample.
P3	Collapse of the mass response	The radius–mass coefficient should decrease from at least 0.4 below the transition to less than 0.05 on the giant branch.	The giant branch retains a substantial positive mass slope, or the low- and high-mass slopes become statistically indistinguishable.
P4	Metallicity sign reversal	The 10–30 $M_\oplus$ population should have a positive conditional metallicity coefficient, while giant planets should have a negative coefficient.	Homogeneous stellar spectroscopy yields the same sign in both mass regimes or eliminates the slope contrast.
P5	Location of the response turnoff	Moving-window or hierarchical varying-coefficient fits should place the irradiation zero crossing and mass-slope collapse within a broad 50–100 $M_\oplus$ interval.	The transition is absent, or its center lies outside approximately 30–150 $M_\oplus$ after selection effects are modeled.
P6	Sufficiency of the giant-planet fold	For giant planets, a model based on $\mathcal{I}_{0.4}$ and metallicity should capture most radius variation; an additional free mass term should provide little held-out improvement. The preferred folding exponent should remain within approximately 0.35–0.55.	Explicit mass substantially improves giant-only prediction, or the preferred exponent moves well outside the stated interval.
P7	Persistence of the cool giant shelf	New low-irradiation transiting and radial-velocity-mass giants should continue to concentrate near 9–10 $R_\oplus$ rather than filling the atlas uniformly.	A larger homogeneous sample turns Zone C into a smooth, featureless continuation of Zones D and E.
P8	Predictive replication	In a future independent sample, a branch-dependent response model should reduce host-grouped radius RMSE by at least 20% relative to one global mass–flux surface.	The piecewise or smoothly varying model provides no meaningful out-of-sample improvement.

A particularly discriminating extension is to replace present-day bolometric flux with reconstructed time-integrated XUV exposure. Atmospheric-loss interpretations predict that this substitution should strengthen the negative low-mass response, whereas giant-planet inflation should remain tied more closely to present-day bolometric forcing. Failure to separate those dependencies would favor a common selection or formation explanation over two distinct evolutionary mechanisms.

7 Physical interpretation and relation to prior work

The response turnover is consistent with a transition between two radius-governing regimes. Below the turnover, radius is strongly mass-sensitive and the negative flux coefficient is compatible with envelope loss, inhibited retention, or correlated survival selection. The localized positive metallicity signal at 10–30 $M_\oplus$ may reflect larger cores, larger primordial envelopes, or greater resistance to atmospheric loss in metal-rich systems [7, 8]. The association is not significant when all planets below 80 $M_\oplus$ are combined, so it should not be interpreted as a universal low-mass metallicity law.

Above the turnover, radius becomes nearly independent of total mass in the aggregate population. Irradiation-driven inflation and heavy-element loading can therefore dominate the residual radius variation. Positive giant-planet irradiation responses and negative metallicity responses are consistent with multivariate studies of hot-Jupiter radii and giant-planet enrichment [3, 9, 10].

The irradiation sign reversal itself has precedent. Weiss et al. [11] fitted opposite incident-flux exponents below and above a substantially higher, visually selected mass boundary. Structural mass–radius studies also identify transitions in the Saturn-mass neighborhood [2, 4, 5]. The distinctive contribution of the atlas is the joint organization of three effects—mass-slope collapse, irradiation sign reversal, and metallicity sign reversal—within one broad response turnoff.

The atlas is not an evolutionary track for individual planets. Migration, cooling, mass loss, tidal heating, formation location, and survey selection can all move or select objects in the underlying control space. The diagram should therefore be interpreted as an empirical population response field whose causal decomposition remains to be established.

8 Limitations

The analysis is subject to several limitations:

- the sample is not corrected for transit detection, mass-measurement follow-up, or publication selection;
- planet and stellar parameters are heterogeneous literature measurements with non-uniform covariance information;
- host metallicities are drawn from multiple spectroscopic scales;
- bolometric incident flux is an imperfect proxy for atmospheric escape, which depends strongly on stellar XUV history;
- ages are incomplete and imprecise for many systems; and
- the parameter-space screen creates an exploratory multiplicity burden, even though validation was performed on the selected response family.

The central result would require substantial revision if a homogeneous, selection-modeled sample removed the low- versus high-mass irradiation contrast, eliminated the metallicity sign reversal, or failed to recover a broad transition in the Saturn/sub-Saturn mass range.

9 Conclusions

The Exoplanet Radius–Response Atlas provides a compact view of the current mass-measured transiting population. Its five empirical zones summarize the progression from compact irradiated planets and sub-Neptunes to cool, warm, and inflated giants. The atlas is generated by a more fundamental

mass–irradiation response field in which three radius sensitivities reorganize across a broad 50–100 $M_\oplus$ transition: the mass coefficient collapses, the irradiation coefficient changes sign, and the metallicity coefficient changes from a localized positive association to a negative giant-planet association.

The diagram is useful because it is both descriptive and falsifiable. It predicts branch-specific irradiation slopes, a metallicity sign reversal, a bounded transition mass, approximate sufficiency of the giant-planet folding coordinate, and measurable out-of-sample gains over a global response surface. Independent recovery of those features in homogeneous, occurrence-aware samples would support the atlas as an H–R-like organizing diagram for exoplanets. Their failure would define how the diagram must be revised.

Data and code availability

The archive snapshot, feature-construction scripts, parameter-space screens, validation outputs, figures, and planet-level atlas table are contained in the accompanying reproducibility bundle. The NASA Exoplanet Archive query was

```
select * from ps where tran_flag=1 and default_flag=1 order by pl_name
```

The query was executed on 24 August 2026.

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